

GYM Application

Title	<u>Gym Application</u>
Description	Gym management systems provide fitness businesses the functionality to manage schedules, memberships, and facilities. The Gym management systems should keep track of these main entities. GYM: (Name, Address, Phone number, Sports Equipments, List of coaches, List of subscriptions). Equipment: (Equipment name (<i>such as treadmill, rowing machine, leg curl, etc.</i>), Equipment code, Quantity of this Equipment in the gym, and Equipment photo (<i>optional</i>)). Customer: (ID, Name, Gender, Address, Phone number, Email, Subscription, List of inbodies). Coach: (ID, Name, Gender, Address, Phone number, Email, Working hours per day (<i>max 10 hours per day</i>)). Subscription: the subscription entity keeps track of subscriptions of users. Each subscription has (Customer ID, Assigned Coach ID, Membership Plan). Membership plan: This entity should keep all details regarding the customer plan such as (Start date, Monthly plan (<i>3 days per week or 6 days per week</i>), number of months registered, price of the membership plan (discounts can be applied when the number of months ≥ 3)). InBody: This entity should keep all details regarding the customer health metrics such as (date of InBody, height (in m), total weight (in kg), body fat mass (in kg), minerals (in kg), total body water (in kg), protein (in kg)).
Constraints	- The coach can have 10 customers maximum at a time. - A customer can only subscribe to one membership plan at a time. - A customer with any membership is allowed to only perform one inbody every month (<i>after 30 days from his registration it is free to do another one</i>).
Deliverables	1. Class diagram for the project (Printed) 2. Java project includes all classes and functions in the description. 3. Documentation that contains: a. Your system description. b. Input and output scenarios.
Bonus	1. GUI 2. Any non-trivial function

Required Functionalities	For Customer: 1. Get his coach info (<i>Name, Phone number, working hours</i>) 2. Display for all the Gym Equipment. 3. Display the customer the membership's plan details. 4. Display the in-body information at a specific date. 5. Display for the user how many kilos need to be reduced according to his body (<i>you can get any calculation through the internet</i>) For Coach: 1. Show a list of all his customers. 2. Get the inbody history of any of his customers. 3. Get all the details of a customer by his name. 4. Show a list of all his female/male customers. For Admin: 1. Can add/edit/delete coaches, types of equipment, and customers to the gym. 2. Show the subscription history for a customer. 3. Display all the customers that subscribed to the gym in a given month/day. 4. Display all the customers of a specific coach. 5. Display the GYM income in a given month. 6. Display the coaches sorted in terms of the most assigned number of customers to the coaches.
Notes:	1. You should implement all concepts of OOP. 2. Each member MUST work on at least one of the required classes besides file processing or GUI. (Individual marks) 3. The evaluation will be mainly based on the student's ability to use and apply OOP concepts and the explanation of the code. 4. You must deliver the Class Diagram for the project. 5. You must apply exception handling. 6. Using Files is mandatory (Not Database) 7. Any project must have at least 8 classes 8. Regarding files: - You must have only two functions for file reading and writing. - You should read data once at the beginning of your run then do your operations and access the code then save in files at the end of your program.